Physical Oceanography (MO8013), Central Concepts 2026

Fredrik Bergelv

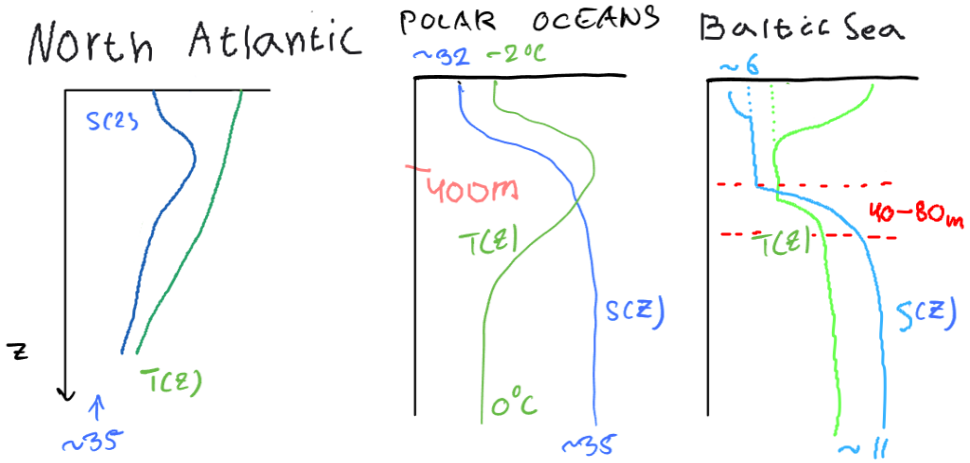
Stockholm University
Department of Meteorology

Physical features of the ocean and relevant dynamical approximations

1. Sea Water: the equation of state $\rho(S,T,P)=\rho_0[1-\beta_T(T-T_0)+\beta_S(S-S_0)+\beta_p(p-p_0)]$. At 25 g/kg, the TMD and freezing point meet. Below we have brakish, above the freezing point is the maximum density.

2. The buoyancy frequency N measures the stability . However the comprassability-term cannot be negelected if one deals with the deep ocean, $N^2 \approx -g\left(\frac{1}{\rho_0}\frac{\partial \rho_0}{\partial z}+\frac{g}{c_s{}^2}\right)\approx -\frac{g}{\rho_0}\frac{\partial \rho_0}{\partial z}$.

3. Sketch vertical T and S profiles in the North Atlantic, the North Pacific, the Arctic Ocean, and the Baltic. Major ocean currents, e.g., the Gulf Stream and Kuroshio.



4. Interpret T –S diagrams: Consider two water masses with different T and S; illustrate in a T–S diagram what happens if the water masses mix. Where can you find the water masses AABW (Antarctic Bottom Water), AAIW (Antarctic Intermediate Water), and NADW (North Atlantic Deep Water)?

5. Write down the hydrostatic dynamical equations on the tangent plane are $\frac{D\vec{u}}{Dt}=\frac{1}{\rho_0}\nabla p-g\hat{z}+\nu\nabla^2\vec{u}$.

6. Hydrostatic applies when the horizontal scales are much larger than vertical scales, Boussinesq applies when the perturbation of density is small.

$$\frac{D\vec{u}_1}{Dt} = -g\nabla_H \eta_1$$

7. In a two layer model we have $\frac{D\vec{u}_2}{Dt} = -\frac{\rho_1}{\rho_2}\left(g\nabla_H \eta_1 + g'\nabla_H \eta_2\right)$, limit gives $\frac{D\vec{u}_1}{Dt} \approx -g'\nabla_H h_1$

$$\frac{\partial \vec{\eta}_i}{\partial t} + h_i\left(\nabla_H \cdot \vec{u}\right) = 0$$

Ekman

1. The equations governing the Ekman flow are $f\,\hat{z}\times\vec{u} = -\frac{1}{\rho_0}\nabla p + \frac{1}{\rho_0}\frac{\partial \vec{\tau}}{\partial z}$.

2. The horizontal Ekman transport is related to the surface wind stress as $\vec{U}_E = -\frac{1}{\rho_0}\hat{z}\times\left(\frac{\vec{\tau}_0}{f}\right)$.

3. The vertical surface Ekman velocity is related to the surface wind stress as $\vec{W}_E = \frac{1}{\rho_0}\nabla_H \times\left(\frac{\vec{\tau}_0}{f}\right)$.

4. In a *cyclonic* circulation bottom Ekman transport → divergence and upward motion.
In an *anticyclonic* circulation bottom Ekman transport → convergence and downward motion

5. Illustrate costal upwelling and downwelling, and equatorial upwelling using simple sketches. Wind driven gyres (Vallis 14)

Wind driven gyres

1. The Sverdrup relation gives $\beta v H = f w_E = \frac{1}{\rho}\nabla\times\vec{\tau}_0$.

2. The Stommel's solution to the wind-driven gyre problem: ad shear stress $\Gamma\frac{\partial v}{\partial x} = \Gamma\frac{\partial^2 \psi}{\partial x^2}$ → $\psi = \psi_{SV}\left[1 - \exp\left(\frac{x}{\delta_s}(x - x_w)\right)\right]$.

Waves: Poincare and Kelvin

1. Given the linearised SWE solution gives Poincare Waves as $\omega^2 = f^2 + c^2 k^2$. Assuming $\eta = \eta_0\cos(kx - \omega t)$ and solving for $\frac{v_0}{u_0} = \frac{f}{\omega}$ → internal oscillations when $k \to 0$.

2. Derive Kelvin waves travelling in the positive x- direction along a coast located at y = 0 by assuming $\eta = \eta_0\exp\left(\frac{i}{k}[x \mp \omega t]\right)$ → $\omega = ck$.

3. The features of an equatorial Kelvin Wave are given by $f \approx \beta y$. The speed is either $c_{BT} = \sqrt{gH}$ or $c_{BC} = \sqrt{g' h_1}$.

4. There is a seiche in the Baltic, extending from St Petersburg to southwestern Baltic (around Lübeck, Germany). Suppose that the seiche is a Kelvin wave. The period is given by $T = 2L/c$, since a seiche is a standing wave.

5. The main lunar tide M_2 has a period of **12.4 hours** and the main solar tide S_2 has period of 12.0 hours.

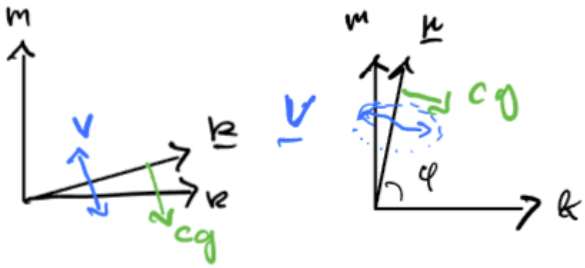
Internal waves and a rotating stratified fluid

These waves are sometimes just referred to internal waves; names including inertial–gravity waves and rotating gravity waves are also used in the literature.

1. The frequency band of internal waves is $N \geq \omega \geq f$, $N \sim 10^{-2}$ s^{-1} and $f \sim 10^{-4}$ s^{-1}. If we have a case where $f > N$ no waves can exsist.

2. The *dispersion relation of internal waves* are detrmined by the aspect ratio (α), which is the ratio between the vertical scale and the horizontal scale. Large α implies $\omega \sim N$; and small α implies $\omega \sim f$.

3. A sketch showing the wavenumber vector, $\vec{k} = (k, m)$, and lines of constant phase for an internal wave with $\omega \sim N$ and $\omega \sim f$. Indicate the phase and group velocities in the diagram.



4. Barotropic tidal currents flowing back and forth over seafloor topography can generate internal waves, also known as internal tides. The internal tides have the same frequency as the barotropic tide. Using the Poincare wave dispersion relation $\omega^2 = f^2 + c^2(k^2 + l^2)$to compute the wave length of the M_2 tide (period 12.4 h) for the barotropic tide (use $c = 200$ ms^{-}1) the internal tide (using $c = 2$ m s^{-1}), if$f = 10^{-4}$ s^{-1}. This gives $\frac{\lambda_{BT}}{\lambda_{InTi}} \approx 100$.

Rossby Waves

Rossby waves are a type of waves that arise from and propagate on a background gradient in potential vorticity. Planetary Rossby waves arise from the latitudinal gradient of f, $\beta = df/dy$. Along the continental slopes in the ocean, there are topographic Rossby waves, propagating along the slope, which arise from the depth (H) variations of the background potential vorticity f/H.

1. On the mid-latitude β-plane the linearised SWE vorticity equation is $\frac{\partial \xi}{\partial t}+f_0 \nabla \cdot u+\beta v=0,$ and the continuity equation is $\frac{\partial \eta}{\partial t}+H_0 \nabla \cdot u=0.$ Using this and assuming a leading order gestrophic baance gives the Rossby wave equation.

2. Assuming a plane wave, $\eta \propto \exp[i(kx+ly-\omega t)]$ gives Rossby wave dispersion relation $\omega=-\frac{\beta k}{L_D^{-2}+k^2+l^2}.$

3. The phase speed is always westward.

4. Long Rossby waves are non-disspersive and move westward.

5. The Rossby wave propagation mechanism is due to the β-plane causing new highs and lows.

6. The maximum frequency of a Rossby waves occurs where $\partial \omega / \partial k=0$, implying that the east–west group velocity is zero: $\omega max=\beta/(2\sqrt{L_D^{-2}+l^2})$; if $l=0$, then $\omega_{max}=L_D \beta/2.$

7. If $l=0$ and $L_D=30$km, the shortest period of Rossby waves at 45 degrees north (or south) is given by $T=4\pi/L_D \beta.$

8. Since ω is conserved in wave reflections, a Rossby wave moving westward will be a long wave with low amplitude, while the rereflected wave will have high amplitude but move far slower (short wave). → Western intensification.

9. how the Stommel–Sverdrup gyre develops from an initial state when the wind is switched on can be understood from Rossby waves. LRW will move to the west, slower SRW will be stuck there and die due to friction.

Baltic Sea

1. An estuary is a semi-enclosed sea, with restricted communication with the ocean and a net supply of freshwater from river run off and precipitation. The Baltic Sea, the Black Sea and the Arctic Ocean are estuaries. The typical circulation in an estuary is fresh water on top, saltier down under with a trong halocline. Ouflow above, inflow below.

2. Knudsen's relations: Consider a two-layer description of an estuary, with upper (lower) layer salinity and volume flow. Conservation of salt and volume gives: $M_1 S_1-M_2 S_2=\rho_f(E-P-R)$ and $M_1 S_1+M_2 S_2=0.$

3. Sketch a typical salinity profile in the central Baltic. Indicate roughly the depth of the halocline, and give typical surface and bottom salinities.

4. Deep-water renewal occurs when sufficiently saline and dense water enters through the Danish Straits and moves into the deeper basins. Major Baltic Inflows are baraotropic inflows due to shift in winds.

5. The Mediterranean Sea is a basin with a negative freshwater balance (evaporation exceeds precipitation plus river run off). The outflow from the Mediterranean Sea through the Gibraltar Strait has a two layer character, with inflow in the upper part and outflow in the lower part.

Surface gravity waves

1. You stand at beach and observe waves with a period of ten second. What is the wavelength of the waves in the open ocean where the depth is 2500 m? Is the wavelength shorter or longer near the beach? [Hint use the dispersion relation for waves in deep water $\omega = \sqrt{gk}$, and check whether the calculated wavelength is small compared to the depth.]

We caclulate $\lambda = \frac{2\pi}{k} = \frac{gT^2}{2\pi}$

2. Torö is a surfing spot south of Stockholm, located 500 km from the Polish coast. On a good surfing day at Torö the wave period is around 7 s (on less good days with smaller waves the period may be around 4–5 s; in the Atlantic typical periods are 10 s or longer). How long time does it take for waves of this period to travel from the Polish coast? Is it the phase or the group velocity that decides the travel time? What is the wavelength of the waves? [Hint use the dispersion relation for waves in deep water $\omega = \sqrt{gk}$].

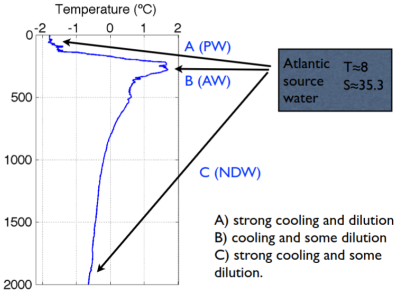
Arctic Ocean

1. An Argo Float measures temperature salinity, depth and time and position.

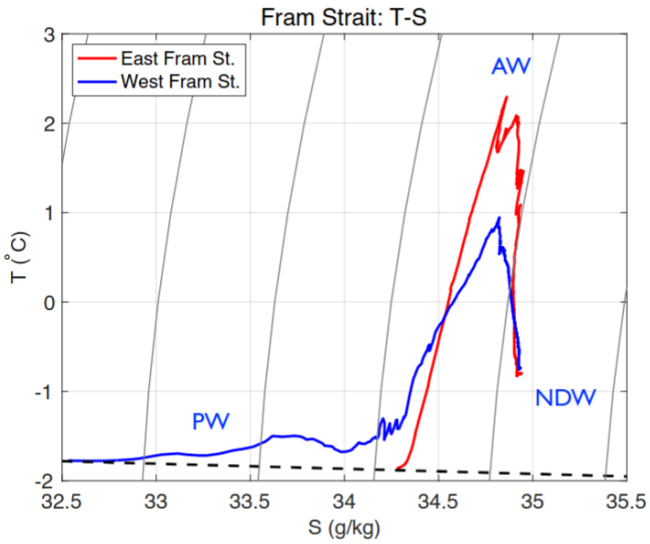
2. How can you estimate the velocity at a point if you know the surface velocity and the horizontal density gradient in the water column? Outline how you would do. Answer: use the thermal wind relation $\frac{\partial u}{\partial z} = \frac{1}{f}\hat{z} \times \nabla b$

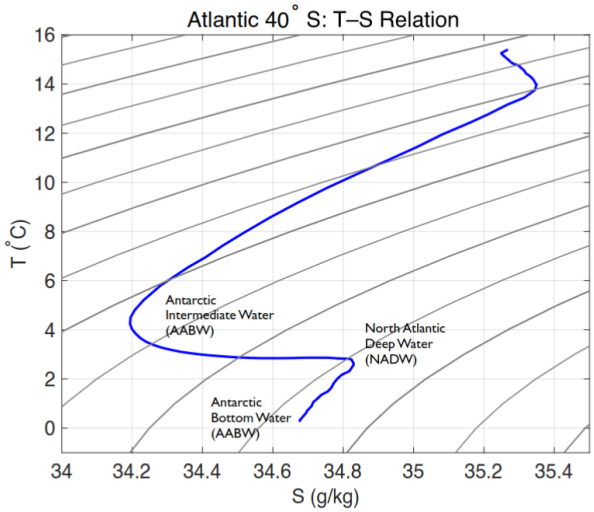
3. Dense water is formed in Nordic Seas by heat loss in the cyclonic circulation, forms cold salt outflow of North Atlantic Deep Water into the Atlantic Basin.

4. Sketch a typical temperature profile in the central Arctic Ocean. Indicate roughly the temperature and depth interval of the Atlantic Water.

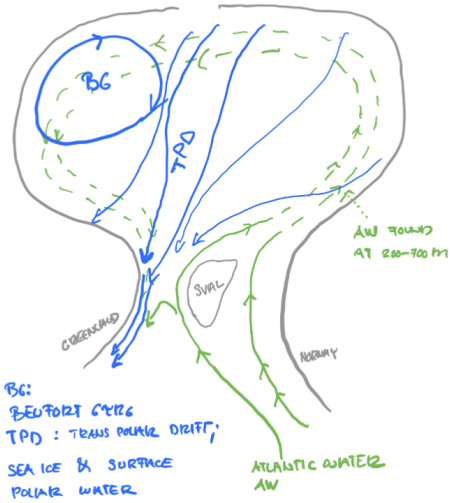


5. Sketch a typical T –S diagram in the central Arctic and Atlantic Ocean. Mark and name three water masses in the diagram.





6. Make a simple diagram of the flow of Atlantic Water and Polar Water in the Arctic Ocean. Indicate where the Atlantic Water is at the sea surface.



7. What characterises the Atlantic Water in the central Arctic Ocean?

Answer: It flows cyclonically – along the continental slope and submarine ridges due to having the same curl as the swind stress. (Derive by taking surf integral of NS and seeing that top and bottom stress give $\oint (\vec{\tau}_t - \vec{\tau}_b) d\vec{S}$, and setting $\vec{\tau}_b = r\,\vec{u}$).

8. Knudsen's relation to the Arctic Ocean.

9. The flow in the Arctic Ocean tends to follow the topography, i.e. the H contours. Consider a barotropic flow in geostrophic balance (take f constant), and use the continuity equation, $\nabla\cdot(\vec{u}h)=0$ which gives $\vec{u}_g\nabla\cdot(h)=0$.

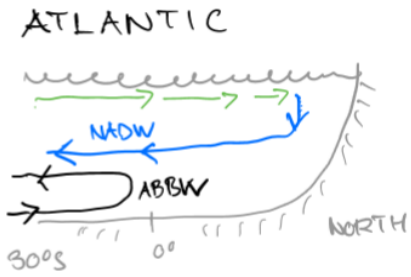
Thermohaline Circulation

The Thermohaline Circulation is a global scale flow pattern that exchanges water between the surface and the deep ocean, and between the ocean basins. It has a long time scale: it takes 100s to 1000s years to renew the deep waters in the World Ocean. This flow is also known as the Meridional Overturning Circulation (MOC); see Vallis chapter 15.

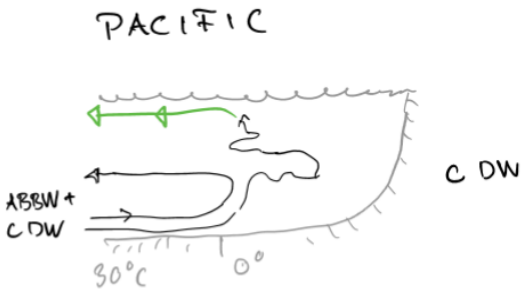
1. In which regions in the World Ocean are deep water formed?
   - Labradour and Nordic seas
   - around Antarctica

2. There is no deep water formed in the North Pacific becouse of the low salinity, so no sinking altough near freezing temperatures.

3. Sketch the the MOC in the Atlantic. Name and mark three important water masses.



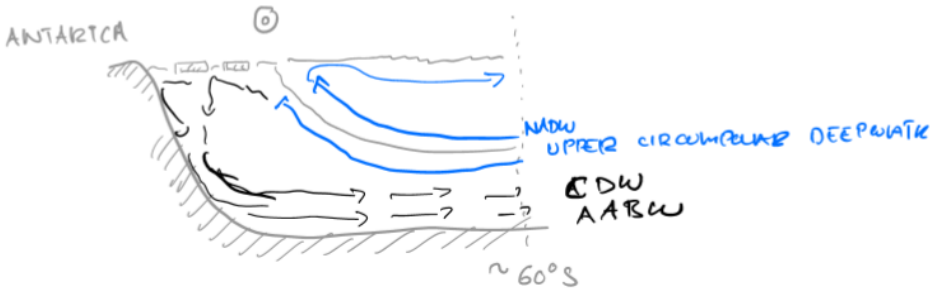
4. Sketch the the MOC in the Pacific.



5. Two processes that drives the MOC are:
   - Surface buoyancy changes due to heating/cooling and freshwater fluxes
   - Wind-driven mixing and upwelling.

5. The oxygen content is low below the surface in the North Pacific since ther is no overturning there.

6. Sketch the the MOC in the vertical–longitudinal in the Southern Ocean. What is the role of the wind forcing and sea ice formation.



7. The observed ocean heat transport is northward at all latitudes (north of the south tip of Africa). How can the Atlantic Meridional Overturning Circulation (AMOC) explain this?

Answer: The AMOC carries warm water northward in the upper ocean and cold water southward below the surface. This cause a net northward heat transport, despite that the winddriven gyre circulations fluxes heat from low to high latitudes in both hemispheres.